Team 5 · Q15 Pitch Script

What should the “you may also like” shelf show, and to whom?

~8 minutes · speak naturally · pause on the charts

Tip: short sentences, look at the chart, then say the one number that matters.

Slide 1 — Title

Hi everyone — we are Team 5.

Our question is:

after someone buys from Lumen Beauty, what should the “you may also like” shelf show — and who does that shelf serve?

We answer this with association rules on one quarter of store data: September to November 2026.

We will move from exploration, to the rules, to one clear recommendation.

Slide 2 — Why the shelf matters

Here is why this question is worth careful work.

Look at this chart: 65% of orders contain only one product.

So in most checkouts, the customer leaves with nothing else in the basket. That empty second slot is where the shelf can help — or fail.

We have 3,401 orders, 647 buyers, and about ¥1.17 million in GMV.

If the shelf helps even 5 to 15% of serum-only orders add a cream, that is roughly ¥2,200 to ¥6,600 per month.

That is only a scenario.

The sharper question for us is this: if we train the shelf badly, who do we leave out?

Slide 3 — Who places the orders?

This chart is central to the “to whom” part of Q15.

Almost half of buyers are light buyers — one or two orders in the quarter.

But they place only about 11% of all orders.

Heavy buyers are 41% of buyers, and they place 80% of orders.

So if we build rules from “everything a customer ever bought,” the algorithm mostly hears the heavy buyers.

Light buyers — almost half of our customers — become hard to see in the data. That is the real risk we wanted to measure.

Slide 4 — Channels and categories

A quick look at the store.

Most orders come from livestream, some from the store shelf.

Serums and creams appear often — they sit at the center of this assortment.

We do not need a long story here. We only need to see that co-purchase is possible, and that serum–cream is a natural place to look.

Slide 5 — Method

Our method is Week 3 association rules — kept clean and honest.

We build baskets in two ways:

1. Order baskets — products bought together in one order. That is the moment the shelf appears.

2. History baskets — everything one customer bought in the whole quarter.

We look only at pairs, because the shelf shows one product next to another.

For every rule we report support, confidence, lift — and also the count and the denominator, so a rare pair is not mistaken for a strong one.

Our thresholds come from the business decision, not from fishing in the data: about 14 orders, or about 13 customers.

And we split buyers into light, mid, and heavy by number of orders.

Slide 6 — Finding: Serum → Cream

Now the finding we trust most.

Serum → Cream.

In 256 of 3,401 orders, a serum and a cream appear together. By chance we would expect about 147. So lift is 1.75. Confidence is 0.40 — four in ten serum orders also have a cream.

Direction matters: cream after serum is stronger than serum after cream.

So the shelf message is clear: after a serum, show a cream.

This is also the only strong category pair. Other pairs at the same bar have lift below 1.

Product-level pairs only help us choose *which* cream — and those rest on just 15 to 27 orders, so we treat them as ranking, not as the main rule.

Importantly, the rule appears in both livestream and store — so it is not only a host pushing a bundle on stream.

Slide 7 — The open slot

This chart shows the opportunity in plain numbers.

Every month, more than 120 serum orders leave without a cream.

Those are customers who already chose a serum — and still did not add a cream.

That is exactly who the shelf can speak to: not with a random bestseller, but with a cream that people actually buy with serum.

Slide 8 — Histories flood the rule table

Here is where the second basket definition becomes important.

At almost the same minimum count, order baskets give 30 rules.

History baskets give 870 — basically every possible pair, all with lift above 1.

Why? Because history baskets are much bigger. A heavy buyer who tried many products creates many pairs, even if those products were never bought on the same day.

And look at the second chart: heavy buyers are 41% of buyers, but they create 93% of the history-rule evidence.

So a history-trained shelf is, in practice, a heavy-buyer shelf.

Slide 9 — History pairs are not “bought together”

We checked the top history pairs inside single orders.

Their history lift looks high — above 2.

But inside the same order, lift falls below 1.

Example: two toners look connected across a customer’s history, but in the same order they appear together only 3 times, when chance expects about 8.

So these are products heavy buyers try on different days — often substitutes, not companions.

If we label that shelf “frequently bought together,” we would be telling a story the data does not support. That distinction is one of the most useful results in this project.

Slide 10 — One shelf for everyone?

So should first-time buyers see a different shelf?

We tested Serum → Cream for light, mid, and heavy buyers, for livestream and store, and on each customer’s first order.

The rule holds everywhere. On first orders, lift is even 1.80.

So our answer is: one shelf is enough — if we build it from orders, not from histories.

A separate first-time shelf is not needed. What first-time buyers need is a shelf that was not trained only on heavy buyers’ long histories.

Slide 11 — Monday morning

Here is what we recommend the merchandising team does next.

1. Mine the shelf from single orders, not whole histories.
2. After any serum, show a cream.
3. Keep one shelf for everyone, including new buyers.
4. Run a clean pilot: from 1 December, show the new shelf to a random half of buyers for six weeks.

Success: at least 5% of serum-only orders add a cream.

Stop if: the add-on rate stays under 2% and not better than control, or light buyers are left behind, or the rule’s lift falls to 1 or below on the control half.

We are not asking the brand to believe a model forever. We are asking them to test one clear idea — and drop it if the evidence says no.

Slide 12 — Close

Three sentences.

Show a cream after a serum.

Build the shelf from orders, not histories.

One shelf for everyone — including people with no history yet.

That is how Lumen Beauty can make the “you may also like” shelf both smarter and fairer.

Thank you — we are happy to take questions.

Timing guide (optional)

Speaking tips for the team

• Point to the chart first, then say one number.

• If stuck: “The simple story is cream after serum, from orders not histories.”

• For Q&A;: support/confidence are frequency filters, not significance tests; the ¥ figure is a scenario; the A/B test is the real measurement.

• Interest comes from the numbers and the logic — not from calling the project exciting.